

Parking Setu **Digital Valet System**

Modernizing five-star hospitality valet operations with instant digital tickets, seamless vehicle tracking, and automated guest pre-retrieval.

ENTERPRISE STUDIO

TenCore Digital Labs Private Limited

Enterprise & Mobile Platform Architecture

LEADERSHIP

Devendra Vaja

Founder & Principal Systems Architect

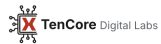
PROPOSAL SPECIFICATION

Fixed-Price MVP (₹2,40,000)

Phase 1 of 5-Phase Strategic Roadmap

Deep-Tech Engineering Leadership

TenCore Digital Labs Private Limited — 24+ Years of Embedded Systems, Kernel Drivers & Android Architecture



Devendra Vaja

Founder & Principal Systems
Architect • TenCore Digital Labs
Private Limited

24+ years of engineering experience across India, USA, and Japan spanning embedded firmware, kernel device drivers, and high-throughput Android mobile platform architecture.

Global Systems Engineering (India • Japan • USA)

24+ years building mission-critical, zero-fault software across India, Tokyo (high-security gaming machine device drivers), and the United States — designing digital solutions built for safety, reliability, and business scale.

High-Throughput Mobile & Offline-First Architecture

Deep-rooted expertise in native Android platforms, low-latency device drivers, local SQLite/Room transactional storage, and resilient background synchronization under spotty network conditions.

Turnkey Enterprise Platforms & Secure IoT Integration

Architecting end-to-end hardware-software bridges, encrypted camera/QR telemetries, and secure cloud workflows with zero-leakage financial and audit integrity.

SECTION 01 • OPERATIONAL CONTEXT

The Hospitality Valet Challenge

Operational friction, bottleneck queues, liability exposure, and the imperative for digital transformation.



Traditional Paper Valet vs. Parking Setu Digital Platform

Transforming manual, error-prone front-of-house friction into a seamless luxury guest experience.

Traditional Paper Valet Friction

LEGACY CHALLENGES

- **Slow Paper Tagging (45–90s):** Hand-writing vehicle slips causes peak-hour arrival bottlenecks at the porch.
- **Lobby Crowding & 15m Wait:** Guests physically stand at the valet desk waiting for staff to locate keys and fetch vehicles.
- **Unverifiable Damage Disputes:** Zero visual proof at check-in leaves the hotel vulnerable to false vehicle damage claims.
- **Key Misplacement Risk:** Manual key-rack tags lead to misplaced fobs, delayed departures, and guest distress.
- **Zero Managerial Telemetry:** Management has zero live visibility into queue length, runner turnaround times, or staff throughput.

Parking Setu Digital Shield

DIGITAL SOLUTION

- **< 5-Second Digital Check-In:** Swift mobile scanning generates an instant digital valet pass via SMS / Web link.
- **1-Tap Pre-Arrival Request:** Guests request their car from the restaurant, lounge, or room before stepping outside.
- **Timestamped Photo Vault:** Instant 4-point vehicle intake photos create an indisputable digital condition record.
- **Precise Bay & Key Tracking:** Every vehicle and key tag is mapped to an exact rack and bay with staff accountability.
- **Live Executive Command Center:** Real-time dashboard tracks turnaround SLAs, runner metrics, and queue alerts.

End-to-End Operational Lifecycle

A frictionless 6-step digital workflow engineered for maximum speed and zero operational gaps.



1. Rapid Ingress & Intake

Valet attendant inputs vehicle number and guest mobile in <5s, captures intake photos, and tags the key fob.



2. Instant Digital Ticket Issuance

Guest instantly receives a secure digital ticket link via SMS (no app download required) with live status.



3. Precision Parking & Bay Logging

Attendant parks the vehicle, records the exact Bay/Zone ('Area → Zone → Bay'), and places key in assigned rack slot.



4. One-Tap Guest Pre-Retrieval

Guest taps 'Request My Vehicle' from dining table or room. Staff runner receives immediate dispatch alert.



5. Swift Retrieval & Staging

Runner locates key and bay instantly, fetches vehicle to porch, and marks it 'Ready' — triggering guest notification.



6. Secure Handover & Audit Close

STEP 1 • ARRIVAL

Intake & Intake Photos



STEP 2 • PASS

Digital Token Generated



STEP 3 • STORAGE

Bay & Key Tag Linked



STEP 4 • REQUEST

Pre-Retrieval Alert



STEP 5 • RETRIEVAL

Vehicle Brought to Porch



Three Role-Specific Digital Touchpoints

Engineered specifically for guests, field valet runners, and hotel management.



Guest Web Experience (PWA)

Zero app install required. Lightweight mobile web ticket with dynamic status tracker, live timer, 1-tap pre-retrieval button, and hotel support link.

Zero App Download Friction



Valet Staff Android App

High-speed native Android tool for attendants. Rapid number plate entry, CameraX damage photo capture, QR scanning, offline queue, and runner dispatch.

< 5s Check-In Turnaround



Hotel Operations Dashboard

Centralized managerial web command center. Live queue monitoring, active inventory, key rack status, runner SLA metrics, audit logs, and exception resolution.

100% Real-Time Visibility

Critical MVP Features • Front-of-House

Built to eliminate guest wait times and protect property reputation.



1. Token-Based Digital Valet Ticket

Secure, tamper-proof QR valet pass sent via SMS. Displays drop-off timestamp, car details, live status, and contact support without exposing raw PII.



2. Key Rack & Slot Tracking

Unique key tag numbers linked directly to storage rack bays. Staff transfers and retrievals are recorded with staff identity and timestamp.



3. Timestamped Vehicle Condition Record

4-Point intake photo capture and notes on existing scratches, odometer, or fuel. Creates an indisputable record preventing false liability claims.



4. Guest Pre-Arrival Retrieval Request

Guests trigger vehicle retrieval in advance with duplicate request protection. Eliminates porch bottleneck crowds during peak checkout and banquet hours.

Critical MVP Features • Operations & Control

Full operational control, staff dispatching, and enterprise governance.



1. Intelligent Runner Dispatch Workflow

Supervisor dashboard highlights pending vehicle requests, displays exact parking bay and key rack slot, and assigns runners with latency tracking.



2. Real-Time SMS & Push Alerts

Automated event-triggered notifications for ticket issuance, vehicle parked confirmation, ready-at-porch alert, and handover completion.



3. Two-Factor Secure Handover

Staff verifies guest via QR token or mobile OTP before vehicle handover. Ensures vehicles and keys are returned exclusively to verified owners.



4. Tamper-Proof Audit & Role Governance

Role-based permissions (Attendant, Supervisor, Admin). Every status change, manual override, or exception is logged with user ID and timestamp.

Architecture, Team & Fixed-Price Model

A defensible ₹2,40,000 fixed price backed by an explicit 120 person-day effort model and modern tech stack.



Modern, Scalable Technology Stack

Engineered with industry-standard frameworks ensuring high throughput, offline resilience, and data security.



Native Android App (Kotlin + Jetpack Compose)

Built with CameraX for rapid damage photo capture, ML Kit for QR scanning, and Room database for local offline operation.

FRONTEND • GUEST / STAFF

Android Kotlin + Next.js PWA



Guest & Admin Web (React / Next.js)

Zero-friction PWA for guests and real-time responsive dashboard for hotel supervisors with live queue telemetry.

API • WORKFLOW LAYER

Spring Boot + REST Engine



Robust Core Backend (Kotlin / Spring Boot)

Enterprise-grade REST APIs, state machine workflows, and OpenAPI/Swagger documentation.

CACHE • QUEUE BUFFER

Redis State & Rate Throttler



PostgreSQL & Redis Infrastructure

PostgreSQL as system of record for audit logs; Redis for OTP throttling, queue state, and duplicate request protection.

DATABASE • AUDIT VAULT

PostgreSQL + S3 Signed Vault

120 Person-Day Effort Model (₹2,40,000 Justification)

Transparent engineering breakdown based on a flat ₹2,000/day blended rate across 120 person-days.

Work Package	Effort (in Days)	Allocated Cost	Key Output Deliverables
Discovery & Technical Design	6	₹12,000	Scope freeze, property workflow schemas & API contract
UX Wireframes & Visual Layouts	7.5	₹15,000	Staff app, guest PWA & hotel dashboard wireframes
Backend, PostgreSQL & Redis	30	₹60,000	REST APIs, auth, state machine & tamper-proof audit vault
Valet Staff Android App	36	₹72,000	Kotlin/Compose native app, CameraX damage photos & Room queue
Guest Mobile PWA	14	₹28,000	QR ticket viewer, pre-request & status tracker
Hotel Operations Dashboard	16	₹32,000	Live queue monitor, runner dispatch & reports
Notifications & Cloud Deployment	5.5	₹11,000	SMS gateway, S3 photo store & cloud deployment
QA, UAT & Staff Training	5	₹10,000	Manual regression, dry runs & on-property staff training
TOTAL MVP EFFORT	120	₹2,40,000	Turnkey 1-Property Deployment (100% Fixed-Price Scope)

Scope Discipline: What Keeps the MVP Affordable

Clear boundaries that ensure turnkey delivery within budget without compromising operational reliability.

Included in Fixed ₹2,40,000 MVP

DELIVERABLES INCLUDED

- **1 Property Deployment:** Full single-hotel property configuration and floor area setup.
- **3 Working Applications:** Native Android Staff App, Guest Web PWA, and Hotel Admin Dashboard.
- **Complete Core Valet Cycle:** Ingress, damage photos, key rack tagging, pre-retrieval, and handover.
- **SMS Gateway Integration:** Automated ticket delivery and status update messages.
- **Complete Source Code & Training:** Full code handover, cloud deployment, and staff training walkthroughs.

Deferred to Phase 2–5 Roadmap

DEFERRED EXPANSIONS

- **PMS / POS Integration:** Hotel room billing and Opera/PMS sync (Phase 3).
- **Payment Gateway / Cashless UPI:** Dynamic parking tariffs and payment collection (Phase 3).
- **Automated ANPR Cameras:** Automatic camera-based license plate recognition (Phase 5).
- **Multi-Property Group SaaS:** Enterprise multi-tenant HQ hierarchy (Phase 4).
- **Third-Party Cloud Usage:** Cloud hosting, SMS/WhatsApp usage billed at actuals.

8–10 Week MVP Delivery & Commercial Milestones

Structured milestone schedule with progress-linked commercial disbursements.

MILESTONE 1 (30%)

Kickoff & Scope Freeze

 **Weeks 1 – 2**  **₹72,000**

Deliverables:

- › Requirement lock & property schema mapping
- › UI/UX wireframes for Staff App & Guest PWA
- › Backend schema and API contract freeze

Deliverable: Architecture & Design Sign-off

MILESTONE 2 (30%)

Core Backend & Staff App

 **Weeks 3 – 5**  **₹72,000**

Deliverables:

- › Spring Boot API, PostgreSQL & Redis setup
- › Android Staff App (Ingress, Photos, Keys)
- › CameraX photo capture & Room offline buffer

Deliverable: Working Staff Sandbox Build

MILESTONE 3 (25%)

Guest PWA & Dashboard

 **Weeks 6 – 7**  **₹60,000**

Deliverables:

- › Guest Web PWA with live status tracking
- › Hotel Management Web Dashboard
- › SMS gateway & real-time notifications

Deliverable: End-to-End System Integration

MILESTONE 4 (15%)

UAT, Training & Go-Live

 **Weeks 8 – 10**  **₹36,000**

Deliverables:

- › On-property dry runs & staff training
- › Production cloud deployment & data backup drill
- › Final documentation & source code handover

Deliverable: Production Handover & Go-Live

Infrastructure & Operating Reserve Model

Transparent estimates for cloud hosting, SMS gateway, and storage requirements.

Initial Infrastructure Reserve

ONE-TIME SETUP

- **Cloud Environment Setup:** AWS / GCP Mumbai region configuration (₹0–₹5,000).
- **Managed PostgreSQL & Redis:** Initial instance provisioning (₹0–₹8,000).
- **Domain, SSL & Security:** Custom property domain & certificates (₹1,000–₹3,000).
- **SMS / OTP Messaging Wallet:** Prepaid SMS credits for pilot (₹5,000–₹10,000).
- **Recommended Total Reserve:** ₹25,000 – ₹50,000 (paid directly to cloud vendors).

Estimated Monthly Operating Cost

MONTHLY OPEX

- **API & Application Hosting:** Scalable cloud instances (₹1,500 – ₹5,000/mo).
- **Managed Database & Storage:** Automated PostgreSQL + S3 photos (₹2,000 – ₹7,000/mo).
- **SMS Messaging Volume:** Scaled by valet guest transaction volume (₹1,000 – ₹8,000+/mo).
- **WhatsApp Business (Optional):** If official business account enabled (₹1,000 – ₹10,000+/mo).
- **Estimated Monthly Total:** ₹6,100 – ₹35,000/mo (at direct actual cost).

SECTION 03 • ROADMAP & EXPANSION

Strategic 5-Phase Roadmap & Support Tiers

From single-property MVP to multi-property enterprise hospitality platform.

Strategic 5-Phase Hospitality Operations Roadmap

A structured 18–30 month evolution scaling from initial pilot validation to group-wide enterprise deployment.

PHASE 1 (MVP)

Operational Core

🕒 8–10 Weeks 📦 ₹2.40 Lakhs

Deliverables:

- › Single hotel validation
- › Android app + Guest PWA + Admin Web
- › Digital tickets, damage photos & audit log

Scope: Core MVP Workflow

PHASE 2

Production Hardening

🕒 6–8 Weeks 📦 Phase 2 Scope

Deliverables:

- › Official WhatsApp Business integration
- › Offline queue conflict resolution
- › Incident & damage claim supervisor approval

Focus: Daily Operational Reliability

PHASE 3

Commercial & Events

🕒 8–10 Weeks 📦 Phase 3 Scope

Deliverables:

- › Payment gateway / UPI integration
- › Banquet & event parking fast-track mode
- › PMS / POS room charging & folio posting

Focus: Revenue & Event Scalability

PHASE 4–5

Multi-Property & Scale

🕒 Multi-Site 📦 Enterprise Scope

Deliverables:

- › Centralized chain HQ command portal
- › Automated ANPR camera license recognition
- › Group-wide BI analytics & data warehouse

Focus: Hotel Chain Enterprise SaaS

Post-Launch Support & Maintenance Packages

Structured SLA tiers ensuring zero downtime and dedicated technical support after the 30-day warranty.

Basic Maintenance

Essential stability & bug fixes

₹8,000

/ month / property

- ✓ Application & database bug fixes
- ✓ Cloud uptime & monitoring review
- ✓ Minor configuration assistance
- ✓ Standard business-hours response
- ✓ Daily automated cloud backup checks

Ideal for steady daily hotel operations

Standard SLA Support

Proactive care & priority assistance

₹15,000

/ month / property

- ✓ Includes everything in Basic tier
- ✓ Monthly proactive database health check
- ✓ Small content & parking zone updates
- ✓ Priority 4-hour critical issue response
- ✓ Dedicated technical support engineer

Most popular for five-star properties

RECOMMENDED

Priority & Event Ops

High-volume & event-ready support

₹25,000

/ month / property

- ✓ Includes everything in Standard tier
- ✓ Banquet / high-traffic event coverage
- ✓ Rapid 1-hour critical response SLA
- ✓ Monthly operational throughput review
- ✓ Quarterly performance optimization

For busy hotels with regular banquets

Operational Impact: Traditional Paper vs. Parking Setu

Direct feature comparison highlighting tangible operational efficiency gains.

Operational Dimension	Traditional Paper Valet	Parking Setu Digital Platform
Intake & Ticket Speed	45 – 90 seconds (manual slip write-up)	< 5 seconds (Digital QR)
Vehicle Retrieval Request	Guest physically waits at porch (10–15 min)	1-Tap Pre-Request from Room / Dining
Vehicle Damage Protection	Zero visual record; frequent disputes	4-Point Timestamped Photo Vault
Key Location Management	Manual pegboard tags; misplacement risk	Digitally Mapped Rack & Bay Tracking
Guest Engagement Channel	Throwaway paper slip	Branded Mobile Pass via SMS / WhatsApp
Managerial Intelligence	Blind operations; paper logbooks	Live SLA Dashboard & Audit Reports

NEXT STEPS & ENGAGEMENT

Let's Modernize Your Valet Operations

TenCore Digital Labs is ready to partner with your property to engineer and deploy the Parking Setu MVP.



FOUNDER & PRINCIPAL SYSTEMS ARCHITECT
Devendra Vaja



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Scan to schedule pilot kickoff